

198. House Robber

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and **it will automatically contact the police if two adjacent houses were broken into on the same night.**

Given an integer array `nums` representing the amount of money of each house, return *the maximum amount of money you can rob tonight **without alerting the police.***

Example 1:

Input: `nums = [1,2,3,1]`

Output: 4

Explanation: Rob house 1 (money = 1) and then rob house 3 (money = 3).
Total amount you can rob = 1 + 3 = 4.

Example 2:

Input: `nums = [2,7,9,3,1]`

Output: 12

Explanation: Rob house 1 (money = 2), rob house 3 (money = 9) and rob house 5 (money = 1).
Total amount you can rob = 2 + 9 + 1 = 12.

416. Partition Equal Subset Sum

Given an integer array `nums`, return `true` if you can partition the array into two subsets such that the sum of the elements in both subsets is equal or `false` otherwise.

Example 1:

Input: `nums = [1,5,11,5]`

Output: `true`

Explanation: The array can be partitioned as [1, 5, 5] and [11].

Example 2:

Input: `nums = [1,2,3,5]`

Output: `false`

Explanation: The array cannot be partitioned into equal sum subsets.

329. Longest Increasing Path in a Matrix

Given an $m \times n$ integers matrix, return *the length of the longest increasing path in matrix*.

From each cell, you can either move in four directions: left, right, up, or down. You **may not** move **diagonally** or move **outside the boundary** (i.e., wrap-around is not allowed).

Example 1:

9	9	4
6	6	8
2	1	1

Diagram illustrating the longest increasing path in the matrix: $[1, 2, 6, 9]$. The path starts at the bottom-middle cell (1), moves left to the bottom-left cell (2), then up to the middle-left cell (6), and finally up to the top-left cell (9).

Input: matrix = `[[9,9,4],[6,6,8],[2,1,1]]`

Output: 4

Explanation: The longest increasing path is `[1, 2, 6, 9]`.

Example 2:

3	4	5
3	2	6
2	2	1

Diagram illustrating the longest increasing path in the matrix: $[3, 4, 5, 6]$. The path starts at the top-left cell (3), moves right to the top-middle cell (4), then right to the top-right cell (5), and finally down to the middle-right cell (6).

Input: matrix = `[[3,4,5],[3,2,6],[2,2,1]]`

Output: 4

Explanation: The longest increasing path is `[3, 4, 5, 6]`. Moving diagonally is not allowed.